Math 210A Notes

Lance Remigio

October 8, 2025

1 Basics

Proposition (Bijection). The map f is a bijection if and only if there exists $g : B \to A$ such that $f \circ g$ is the identity map on B and $g \circ f$ is the identity map on A.

Definition (Binary Relations, Equivalence Relations, Equivalence Classes, Partitions).

(1) A **binary relation** on a set A is a subset R of $A \times A$ and we write $a \sim b$ if $(a, b) \in R$.

(2) The relation $\sim$ on A is said to be:

   (a) **Reflexive** if $a \sim a$ for all $a \in A$;

   (b) **Symmetric** if $a \sim b$, then $b \sim a$ for all $a, b \in A$;

   (c) **Transitive** if $a \sim b$ and $b \sim c$ implies $a \sim c$ for all $a, b, c \in A$.

Proposition (Isomorphism Properties). If $\varphi : G \to H$ is an isomorphism, then

(a) $|G| = |H|$

(b) G is abelian if and only if H is abelian.

(c) for all $x \in G$, $|x| = |\varphi(x)|$.

2 Group Actions

Definition (Group Actions). A **group action** of a group G on a set A is a map $G \times G \to A$ (written $g \cdot a$ for all $g \in G$ and $a \in A$) satisfying the following properties:

(1) $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$ for all $g_1, g_2 \in G$ and $a \in A$;

(2) $1 \cdot a = a$ for all $a \in A$.

Example (Group Actions).

(1) The additive group $\mathbb{R}$ that acts on $\mathbb{R}^2$ by $r \cdot (x, y) = (x + ry, y)$.

(2) Let G be any group and let $A = G$. The map $g \cdot a : G \times A \to G$ defined by

$$g \cdot a = gag^{-1} \ \forall g, a \in G$$

satisfy the axioms of a group action.

2.1 Permutation Representation of a Group Action

Let the group G act on the set A. For each fixed $g \in G$, the map $\sigma_g : A \to A$ is a

(1) Permutation for each fixed $g \in G$ and

(2) The map from G to S_A defined by $g \mapsto \sigma_g$ is a homomorphism; that is, the map $\varphi : G \to S_A$ is any homomorphism from a group G to a symmetric group S_A defined by

$$g \cdot a = \varphi(g)(a).$$

3 Section 2.1: Subgroups

Definition (Subgroup). Let G be a group. The subset G of G is a **subgroup** of G if $H \neq \emptyset$ and H is closed under products and inverses; that is, for all $x, y \in H$ implies $x^{-1} \in H$ and $xy \in H$. If H is a subgroup of G, we shall write $H \leq G$.

Definition (The Subgroup Criterion). A subset H of a group G is a subgroup if and only if

(1) $H \neq \emptyset$, and

(2) For all $x, y \in H$, $xy^{-1} \in H$.

4 Section 2.2: Centralizers, Normalizers, Stabilizers, and Kernels

4.1 Centralizers, Centers, and Normalizers

Definition (Centralizer). Define $C_G(A) = \{g \in G : gag^{-1} = a \ \forall a \in A\}$. This subset of G is called the **centralizer** of A in G. Since $gag^{-1} = a$ if and only if $ga = ag$, $C_G(A)$ is the set of elements of G which commute with every element of A.

Definition (Center). Define $Z(G) = \{g \in G : gx = xg \ \forall x \in G\}$, the set of elements commuting with all elements of G. This subset of G is called the **center** of G.

Definition (Normalizer). Define $gAg^{-1} = \{gag^{-1} : a \in A\}$. Define the **normalizer** of A in G to be the set $N_G(A) = \{g \in G : gAg^{-1} = A\}$.

- If G is abelian if and only if $Z(G) = G$.
- G is abelian if and only if $C_G(A) = N_G(A)$ for any subset A of G.

4.2 Stabilizers and Kernels of Group Actions

In many cases, we can make our task of proving certain subsets of a group by making it about a problem about whether that subset is equal to either its stabilizer or kernel of a group action. The three main sets mentioned above are just special cases of group actions acting on either a fixed element of some set S or keeping all the elements of S fixed.

Definition (Stablizer). If G is a group acting on a set S and s is some fixed element of S, the **stabilizer** of s in G is the set

$$G_s = \{g \in G : g \cdot s = s\}.$$

Definition (Kernel). Let G be a group acting on a set S. Then the **kernel** of the action of G on S is the set

$$\{g \in G : g \cdot s = s \ \forall s \in S\}.$$

Let $S = \mathcal{P}(G)$ (the power set of G) (the collection of all subsets of G, and let G act on S by *conjugation*), that is, for each $g \in G$ and each $B \subseteq G$ let

$$g : B \to gBg^{-1} \text{ where } gBg^{-1} = \{gbg^{-1} : b \in B\}.$$

This action gives us the stabilizer of A in G; that is,

$$\begin{aligned} N_G(A) &= \{g \in G : gAg^{-1} = A\} \\ &= \{g \in G : g \cdot A = A\} \\ &= G_A \end{aligned}$$

which is clearly a subgroup of G.

5 Cyclic Groups and Cyclic Subgroups

Definition (Cyclic). A group H is **cyclic** if H can be generated by a single element; that is, there exists some element $x \in H$ such that $H = \{x^n : n \in \mathbb{Z}\}$ (where as usual the operation is multiplication).

- If the group in question is additive then we say that $H \subseteq G$ is cyclic if

$$H = \{nx : n \in \mathbb{Z}\}$$

- We say that H is generated by x (or x is the generator of H).
- A cyclic group may have more than one generator.

Proposition. If $H = \langle x \rangle$, then $|H| = |x|$ (where if one side of this equality is infinite, so is the other). More specifically,

(1) If $|H| = n < \infty$, then $x^n = 1$ and $1, x, x^2, \ldots, x^{n-1}$ are all distinct elements of H, and

(2) If $|H| = \infty$, then $x^n \neq 1$ for all $n \neq 0$ and $x^a \neq x^b$ for all $a \neq b$ in $\mathbb{Z}$.

Proposition ((3)). Let G be an arbitrary group, $x \in G$ and let $m, n \in \mathbb{Z}$. If $x^n = 1$ and $x^m = 1$, then $x^d = 1$, where $d = (m, n)$. In particular, if $x^m = 1$ for some $m \in \mathbb{Z}$, then $|x| \mid m$.

Theorem ((4)). Any two cyclic groups of the same order are isomorphic. More specifically,

(1) if $n \in \mathbb{Z}^+$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n, then the map

$$\begin{aligned} \varphi : \langle x \rangle &\to \langle y \rangle \\ x^k &\mapsto y^k \end{aligned}$$

is well defined and is an isomorphism.

Proposition ((5))**.** Let G be a group, let $x \in G$ and let $a \in \mathbb{Z} - \{0\}$.

(1) If $|x| = \infty$, then $|x^a| = \infty$.

(2) If $|x| = n < \infty$, then $|x^a| = \dfrac{n}{(n,a)}$.

(3) In particular, if $|x| = n < \infty$ and a is a positive integer dividing n, then $|x^a| = \frac{n}{a}$.

Proposition ((6))**.** Let $H = \langle x \rangle$.

(1) Assume $|x| = \infty$. Then $H = \langle x^a \rangle$ if and only if $a = \pm 1$.

(2) Assume $|x| = n < \infty$. Then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$. In particular, the number of generators of a H is $\varphi(n)$ (where φ is Euler's φ-function).

This is mainly used to give a description of all the subgroups of H that generate H.

Example (Multiplicative Group)**.** Find all the subgroups of $Z_{20} = \langle x \rangle$.

(1) Find all the divisors of 20. That is, the divisors of 20 are

$$\{1, 2, 4, 5, 10, 20\}.$$

(2) From the above proposition (2) the divisors of 20 gives us the unique subgroups of $\langle x \rangle$ that generate Z_{20}. These subgroups being

$$\langle x \rangle, \langle x^2 \rangle, \langle x^4 \rangle, \langle x^5 \rangle, \langle x^{10} \rangle, \langle x^{20} \rangle.$$

(3) Using prop (5), we can find the order of each subgroup:

$$\begin{aligned}
|\langle x^2 \rangle| &= \frac{20}{(2,20)} = \frac{20}{2} = 10 \text{ elements in } \langle x^2 \rangle \\
|\langle x^4 \rangle| &= \frac{20}{(4,20)} = \frac{20}{4} = 5 \text{ elements in } \langle x^4 \rangle \\
|\langle x^5 \rangle| &= \frac{20}{(5,20)} = \frac{20}{5} = 4 \text{ elements in } \langle x^5 \rangle \\
|\langle x^{10} \rangle| &= \frac{20}{(10,20)} = \frac{20}{10} = 2 \text{ elements in } \langle x^{10} \rangle \\
|\langle x^{20} \rangle| &= \frac{20}{(20,20)} = \frac{20}{20} = 1 \text{ element in } \langle x^{20} \rangle
\end{aligned}$$

(4) List out the generators of each subgroup by computing the Euler-Totient function $\varphi(n)$. For example, given the cyclic subgroup $\langle x^2 \rangle$ which has order 10, we have

$$\varphi(10) = \varphi(5 \cdot 2) = 4 \text{ generators.}$$

Note that φ gives us the number of $a \in \mathbb{Z}^+$ up to 10 such that $(a, 10) = 1$ (where a and 10 are relatively prime). Indeed, we have

- $(1, 10) = 1 \Longrightarrow (x^2)^1 = x^2$ is a generator;
- $(2, 10) = 2 \Longrightarrow (x^2)^2 = x^4$ is NOT a generator;
- $(3, 10) = 1 \Longrightarrow (x^2)^3 = x^6$ is a generator;
- $(4, 10) = 2 \Longrightarrow (x^2)^4 = x^8$ is NOT a generator;

- $(5, 10) = 5 \implies (x^2)^5 = x^{10}$ is NOT a generator;
- $(6, 10) = 2 \implies (x^2)^6 = x^{12}$ is NOT a generator;
- $(7, 10) = 1 \implies (x^2)^7 = x^{14}$ is a generator;
- $(8, 10) = 2 \implies (x^2)^8 = x^{16}$ is NOT a generator;
- $(9, 10) = 1 \implies (x^2)^9 = x^{18}$ is a generator;
- $(10, 10) = 10 \implies (x^2)^{10} = x^{20}$ is NOT a generator;

(5) To describe the containments of each of these subgroups in relation to each other, we fix each divisor in our list and identify which ones can be divided by it. For instance, we see that

$$\begin{aligned} 1 &\mid 2, 4, 5, 10, 20 \\ 2 &\mid 4, 10, 20 \\ 4 &\mid 20 \\ 5 &\mid 10, 20 \\ 10 &\mid 20. \end{aligned}$$

This gives us the following containments:

$$\begin{aligned} &\langle x^{20} \rangle \subseteq \langle x^{10} \rangle \subseteq \langle x^5 \rangle \subseteq \langle x \rangle \\ &\langle x^{20} \rangle \subseteq \langle x^{10} \rangle \subseteq \langle x \rangle \\ &\langle x^{20} \rangle \subseteq \langle x^4 \rangle \subseteq \langle x^2 \rangle \subseteq \langle x \rangle \\ &\langle x^{20} \rangle \subseteq \langle x^2 \rangle \subseteq \langle x \rangle \\ &\langle x^{20} \rangle \subseteq \langle x \rangle. \end{aligned}$$

Theorem. Let $H = \langle x \rangle$ be a cyclic group.

(1) Every subgroup of H is cyclic. More precisely, if $K \leq H$, then either $K = \{1\}$ or $K = \langle x^d \rangle$, where d is the smallest positive integer such that $x^d \in K$.

(2) If $|H| = \infty$, then for any distinct nonnegative integers a and b, $\langle x^a \rangle \neq \langle x^b \rangle$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{|m|} \rangle$, where $|m|$ denotes the absolute value of m, so that the nontrivial subgroups of H correspond with the integers $1, 2, 3, \ldots$.

(3) If $|H| = n < \infty$, then for each positive integer a dividing n there is a unique subgroup of H of order a. This subgroup is the cyclic group $\langle x^d \rangle$, where $d = \frac{n}{a}$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{(n,m)} \rangle$ so that the subgroups of H correspond bijectively with the positive divisors of n.

6 Week 6

6.1 Cyclic

Proposition (8)**.** If $\mathcal{A}$ is any collection of subgroups of a group G, then $\bigcap_{H \in \mathcal{A}} H$ is a subgroup of G.

Proof. This can be shown pretty easily. ■

Definition (The Generating Set of a Group)**.** If A is any subset of the group G define

$$\langle A \rangle = \bigcap_{A \subseteq H, H \leq G} H$$

where A is called the **generating set**.

Notice that in the notation of the above proposition, we have

$$\mathcal{A} = \{H \leq G : A \subseteq H\}.$$

Notice that $\mathcal{A} \neq \emptyset$ as G is an element of $\mathcal{A}$ since $G \leq G$ and $A \subseteq G$.

Proposition. $\langle A \rangle$ is the unique minimal element of $\mathcal{A}$.

Proof. Since $A \subseteq H$ for all $H \in \mathcal{A}$, $A \subseteq \langle A \rangle$. Then $\langle A \rangle \in \mathcal{A}$. Let $K \in \mathcal{A}$. Note that

$$\bigcap_{H \in \mathcal{A}} H \leq K.$$

That is, $\langle A \rangle \leq K$. Hence, $\langle A \rangle$ is the minimal element with respect to inclusion. ■

When A is finite, that is,

$$A = \{\alpha_1, \alpha_2, \ldots, \alpha_n\} \text{ for some } n \in \mathbb{N},$$

we write

$$\langle A \rangle = \langle a_1, a_2, \ldots, a_n \rangle.$$

This is a more concrete version of the previous set $\langle A \rangle = \bigcap_{H \leq G, K \subseteq H} H$. Define the following set

$$\overline{A} = \{a_1^{\varepsilon_1} a_2^{\varepsilon_2} a_3^{\varepsilon_3} \ldots a_n^{\varepsilon_n} : n \in \mathbb{N}, \varepsilon_i = \pm 1 \ \forall i\}.$$

If $A = \emptyset$, define $\overline{A} = \{1\}$.

Proposition. $\langle A \rangle = \overline{A}$

6.2 Quotient Groups and Homomorphisms

Let G be a group and $N \leq G$. Define a relation on G by

$$a \sim b \iff a^{-1} b \in N.$$

It is straightforward to prove that this defines an equivalence on relation on G. For $a \in G$, the equivalence class of a is

$$\begin{aligned}
\{b \in G : a \sim b\} &= \{b \in G : a^{-1}b \in N\} \\
&= \{b \in G : a^{-1}b = n \ \text{ for some } n \in N\} \\
&= \{b \in G : b = an \text{ for some } n \in N\} \\
&= \{an : n \in N\} \\
&:= aN.
\end{aligned}$$

Definition (Cosets)**.** Suppose $N \leq G$ where G is a group and $g \in G$. Let

$$gN = \{gn : n \in N\}$$
$$Ng = \{ng : n \in N\}$$

be called a **left coset** and **right coset** of N in G, respectively. Any element of a coset is called a **representative** of a coset.

We will denote the set of all left cosets of N (or right cosets) in G by G/N (read as G mod N).

Proposition (4). Let $N \leq G$ be a group and G/N forms a partition of G. For all $a, b \in G$, $aN = bN$ if and only if $a^{-1}b \in N$. In particular, $aN = bN$ if and only if a and b are representatives of the same coset.

Proof. Since we have recognized left cosets as equivalence classes induced by an equivalence relation, they form a partition, that is,

$$G = \bigcup_{g \in G} gN$$

and $g_1N = g_2N$ if and only if $g_1N \cap g_2N \neq \emptyset$ for all $g_1, g_2 \in G$.

($\Longrightarrow$) Suppose $a^{-1}b \in N$, then

$$a^{-1}b = n \text{ for some } n \in N.$$

It follows that $b = an \in aN$. Since N is a subgroup, $1 \in N$. Hence, $b \cdot 1 \in bN$. It follows that

$$aN \cap bN \neq \emptyset \Longrightarrow aN = bN.$$

($\Longleftarrow$) Suppose that $aN = bN$. Then $an = b$ for some $n \in N$. Hence, $n = a^{-1}b \in N$. We have

$$\begin{aligned} an = bn &\Longleftrightarrow a^{-1}b \in N \\ &\Longleftrightarrow b \in aN \text{ and } a \in aN \\ &\Longleftrightarrow a \text{ and } b \text{ are representatives of } aN \text{ (or } bN\text{)}. \end{aligned}$$

■

Proposition (5). (1) The operation on G/N described by $aN \cdot bN = (ab)N$ for all $a, b \in G$ is well defined if and only if $gng^{-1} \in N$ for all $g \in G$ and $n \in N$.

(2) If the operation above is well-defined, then G/N forms a group, where $1N$ is the identity and $(gN)^{-1} = g^{-1}N$.

Proof. (1) ($\Longleftarrow$) (We will show that the operation defined above is well-defined) To show that the operation is well-defined, we need to show that

$$(ab)N = (a_1b_1)N.$$

Suppose that $gng^{-1} \in N$ for all $g \in G$ and $n \in N$. Let $a, a_1 \in aN$ and $b, b_1 \in bN$. Then $a_1 = an$ and $b_1 = bm$ for some $n, m \in N$. To get our result, we need to show that $a_1b_1 \in (ab)N$ if and only if $(a_1b_1)N = (ab)N$.

Now, we will prove the former. Indeed, we have

$$\begin{aligned} a_1b_1 &= (an)(bm) \\ &= a(bb^{-1})nbm \\ &= ab(b^{-1}nb)m. && (b^{-1}n(b^{-1})^{-1} \in N)) \end{aligned}$$

Hence, it follows that $a_1b_1 = (ab)n_1m$ where $n_1 \in N$ and $n_1 = b^{-1}n(b^{-1})^{-1}$.

($\Longrightarrow$) This direction is left for the reader to prove.

(2) This is left for the reader to prove.

■

7 Week 7

Last time we constructed cosets from an equivalence relation and looked at the set of all left cosets of subgroup N of a group G, denoted by

$$G/N = \{gN : g \in G\}.$$

We showed that the operation on G/H defined by

$$(aH)(bH) = (ab)N$$

is well-defined when $gng^{-1} \in H$ for all $g \in G$ and for all $h \in N$. This makes G/N a group. We will prove that this operation is associative, the rest is left to the reader.

Let $N \leq G$ a group where "coset multiplication" is well-defined. Let $aN, bN, cN \in G/N$ $(a, b, c \in G)$. Then

$$\begin{aligned} aN(bNcN) &= aN((bc)N) \\ &= (a(bc))N \\ &= ((ab)c)N \\ &= ((ab)N)cN \\ &= (aNbN)cN. \end{aligned}$$

Similarly, we can show that $1N$ is the identity of G/N and show that

$$(aN)^{-1} = a^{-1}N \quad \forall aN \in G/N.$$

Indeed, for every $gN \in G/N$

$$gN = (1_G g)N = (1_G N) \cdot (gN)$$

and

$$gN = (g 1_G)N = (1_G N) \cdot (gN).$$

Now, we have that since $g \in G$ and $g^{-1} \in G$, it follows that

$$1_G N = (gg^{-1})N = (gN) \cdot (g^{-1}N)$$

and

$$1_G N = (g^{-1}g)N = (g^{-1}N) \cdot (gN).$$

This tells us that G/N is a group where N has the nice property of being "normal". This will be expanded in the following definition:

Definition (Normal Subgroups of G). A subgroup N of G is called **normal** in G if every element of g normalizes N. That is, N is normal in G if

$$gNg^{-1} = N \quad \forall g \in G.$$

If N is a normal subgroup of G, then we write $N \trianglelefteq G$.

Theorem (6). Let N be a subgroup of G. The following are equivalent

(1) $N \trianglelefteq G$

(2) $N_G(N) = G$

(3) $gN = Ng$ for all $g \in G$

(4) The operation coset multiplication is well-defined.

(5) $gNg^{-1} \subset N$ for all $g \in G$.

Why does (5) only need containment? It is because you can construct a bijective map between gNg^{-1} and N.

Example (Checking if $\langle s \rangle$ is normal in D_16). We need to examine gsg^{-1} for an arbitrary $g \in D_{16}$. Note that

$$D_{16} = \{1, r, \ldots, r^7, s, sr, \ldots, sr^7\}.$$

Letting $g = s^i r^j$ where $i \in \{0,1\}$ and $j \in \{0,1,2,\ldots,7\}$. Then we have

$$\begin{aligned} gsg^{-1} &= (s^i r^j)s(s^i r^j)^{-1} && (s^i = s^{-i}) \\ &= s^i r^j s r^{-j} s^i \end{aligned}$$

Note that when $i = 0$, we have

$$\begin{aligned} gsg^{-1} &= r^j s r^{-j} \\ &= r^j r^j s \\ &= r^{2j} s. \end{aligned}$$

When $j = 1$, this gives us $gsg^{-1} = r^2 s$. But note that $r^2 s \notin \langle s \rangle$ because otherwise r^2 is either the identity or s which r^2 is neither of these two which is a contradiction. Hence, gsg^{-1} cannot be in $\langle s \rangle$ and so it cannot be a normal subgroup of D_16.

Theorem (Big!)**.** A subgroup N of a group G is normal if and only if it is the kernel of some homomorphism.

Proof. ($\Longrightarrow$) Suppose $N \trianglelefteq G$. Let's define the mapping $\pi : G \to G/N$ by $\pi(g) = gN$ for all $g \in G$. Notice that since N is a normal subgroup of G, this map is well-defined. Let $g_1, g_2 \in G$. Then

$$\pi(g_1 g_2) = (g_1 g_2)N = (g_1 N)(g_2 N) = \pi(g_1)\pi(g_2).$$

Hence, π is a homomorphism. It remains to show that $\ker(\pi) = N$. Indeed, we have that

$$\begin{aligned} \ker(\pi) &= \{g \in G : \pi(g) = 1N\} \\ &= \{g \in G : gN = 1N\} \\ &= \{g \in G : g \in 1N\} \\ &= \{g \in G : g \in N\} \\ &= N. \end{aligned}$$

Hence, $\ker(\pi) = N$.

($\Longleftarrow$) This direction will be on homework. ∎

Definition (Natural Projection)**.** Let N be normal in G. The homomorphism π from the previous theorem is called the **natural projection** (homomorphism) of G onto G/N. If $\overline{G} \leq G/N$, the complete pre-image of $\overline{H}$ is $\pi^{-1}(\overline{H})$.

Remark. If $\overline{H} \subseteq G/N$, then $N \leq \pi^{-1}(\overline{H})$ since

$$1N \in \overline{H} \Longrightarrow N = \ker(\pi) = \pi^{-1}(1N) \subseteq \pi^{-1}(\overline{H}).$$